

Retail Store Product Sales

1. Introduction

The goal of this analysis is to explore a retail sales dataset and identify which variables have the strongest influence on key business outcomes. The dataset contains 15,000 observations with 10 continuous variables covering pricing, promotions, advertising, inventory, weather, and customer behaviour. There are no missing values.

The analysis follows four steps: a correlation analysis to understand how variables relate to each other, PCA to find the main axes of variation in the data, a linear regression to quantify what drives footfall, and a logistic regression to identify what increases the risk of above-average return rates.

2. KPI Selection & Variable Distributions

I selected two KPIs that represent the most actionable business outcomes. The charts below show their distributions in the dataset.

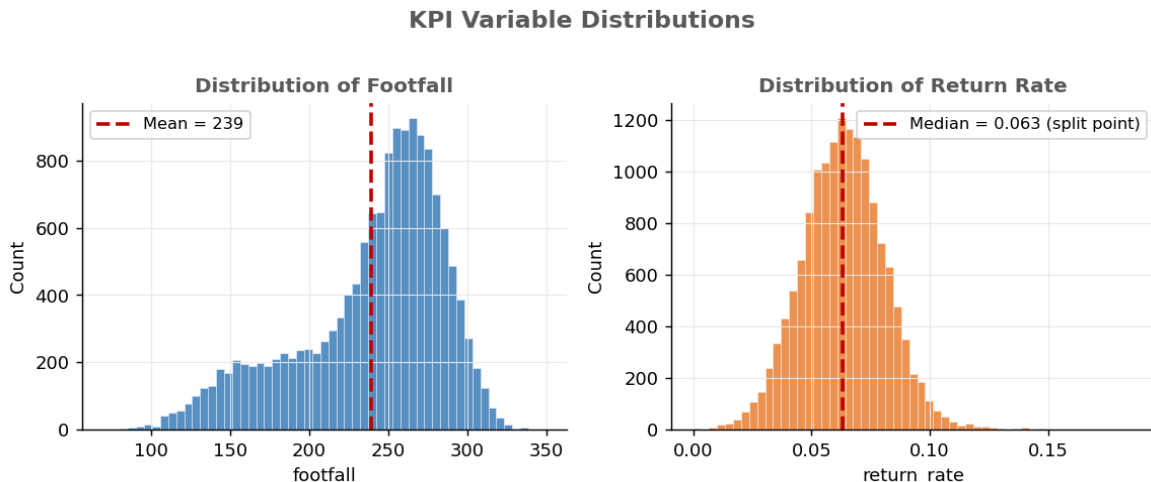


Figure 1 - Distributions of the two target variables. The median return rate (dashed line) is the split point for the binary logistic target.

Footfall is chosen as the linear regression target. It measures daily customer traffic and is driven by controllable inputs like advertising and promotions.

High Return is the logistic regression target. I binarised return_rate at its median to create a balanced 50/50 binary outcome where 1 means above-average return rate. Return rates eat directly into margin, so identifying what conditions produce them is useful.

3. Feature Engineering

Before modelling, I noticed that price and competitor_price have a correlation of 0.969, almost the same variable. Including both in a regression would cause severe multicollinearity. I replaced them with one engineered variable:

$$\text{price_gap} = \text{price} - \text{competitor_price}$$

A positive value means the store is priced above its competitor. I confirmed the fix worked by checking VIFs after fitting the regression, all values were below 5.

4. Correlation Analysis

The correlation heatmap below shows how all nine modelling variables relate to each other. The most important patterns are highlighted.

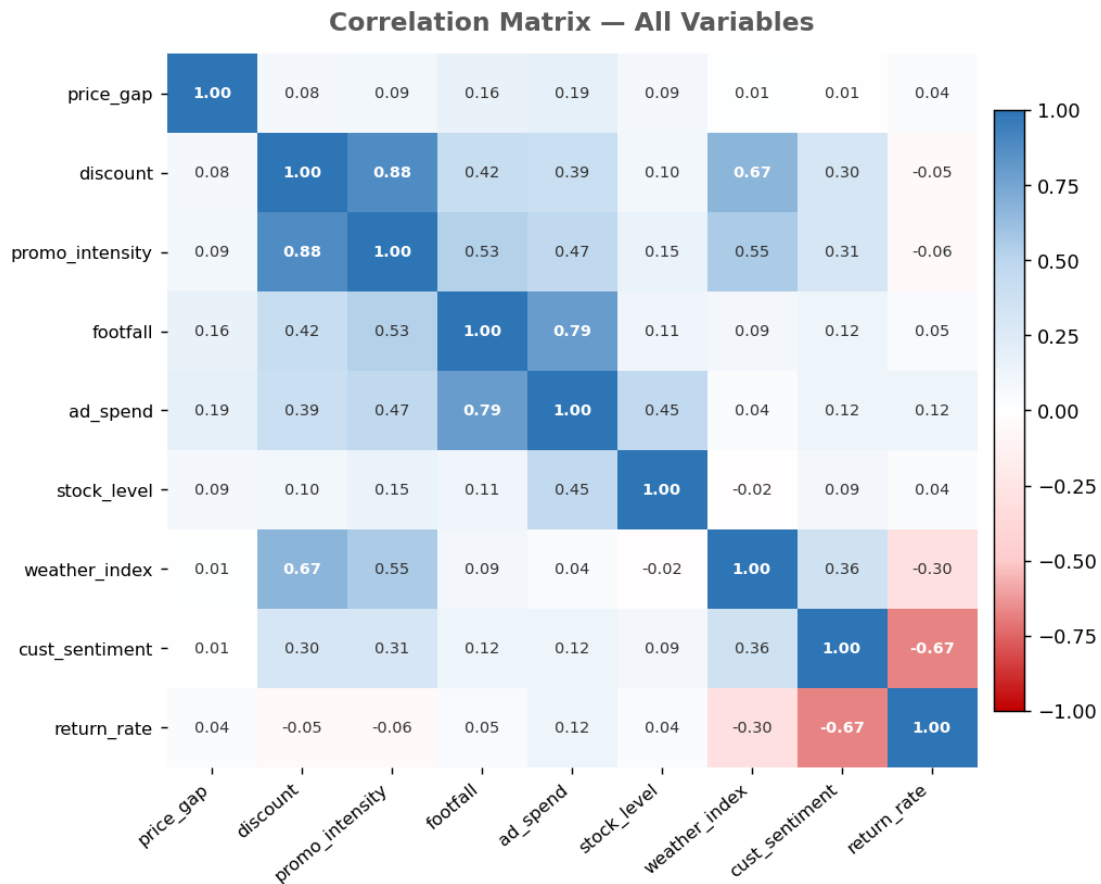


Figure 2 - Pearson correlation matrix. Deep blue = strong positive, deep red = strong negative.

- ad_spend and footfall are strongly correlated ($r = 0.789$), suggesting advertising directly drives customer traffic.
- discount and promotion_intensity move together ($r = 0.881$), so they capture the same underlying phenomenon.
- customer_sentiment and return_rate show the strongest negative relationship ($r = -0.673$) higher satisfaction means fewer returns.
- price and competitor_price are $r = 0.969$, confirming they cannot both enter a model without causing multicollinearity.

The scatter plots below zoom in on the two strongest relationships with the KPI variables.

Key Bivariate Relationships

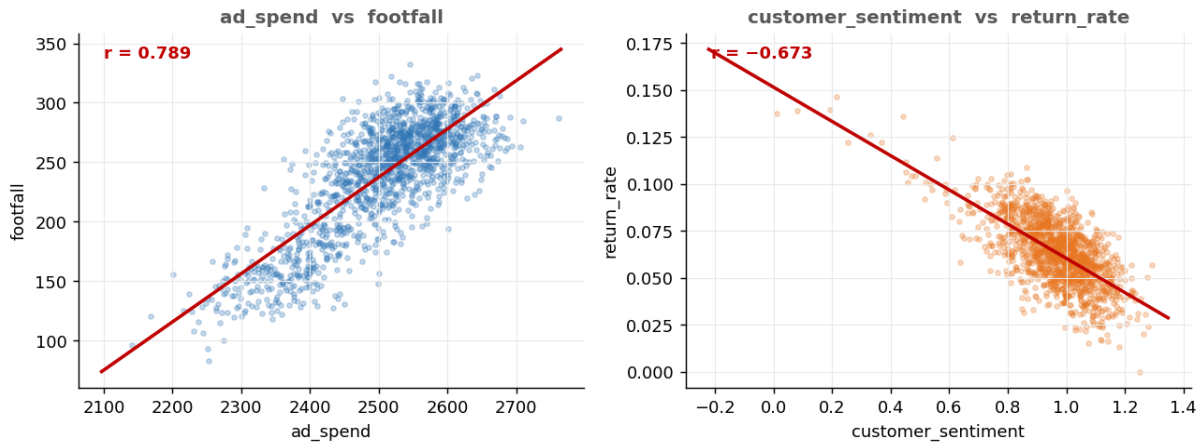


Figure 3 - Left: *ad_spend* vs *footfall* ($r = 0.789$). Right: *customer_sentiment* vs *return_rate* ($r = -0.673$).

5. Principal Component Analysis

I ran PCA on all 10 original variables to understand the underlying structure before modelling. I kept both price and competitor_price here intentionally as their near-identical loadings visually confirm they carry the same information.

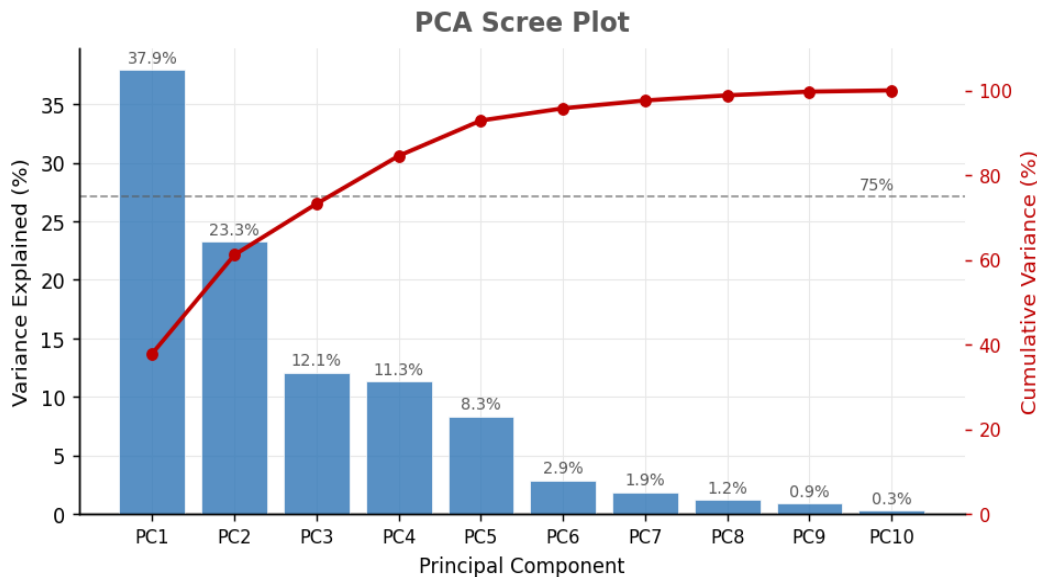


Figure 4 - Scree plot. The first four components explain ~75% of total variance.

Component	Variance %	Cumulative	Main variables driving it
PC1	~28%	~28%	price & competitor_price (+) vs discount & promotion (-)
PC2	~19%	~47%	footfall (+) and ad_spend (+)
PC3	~16%	~63%	weather_index (+) and discount (+)
PC4	~12%	~75%	return_rate (+) and customer_sentiment (-)

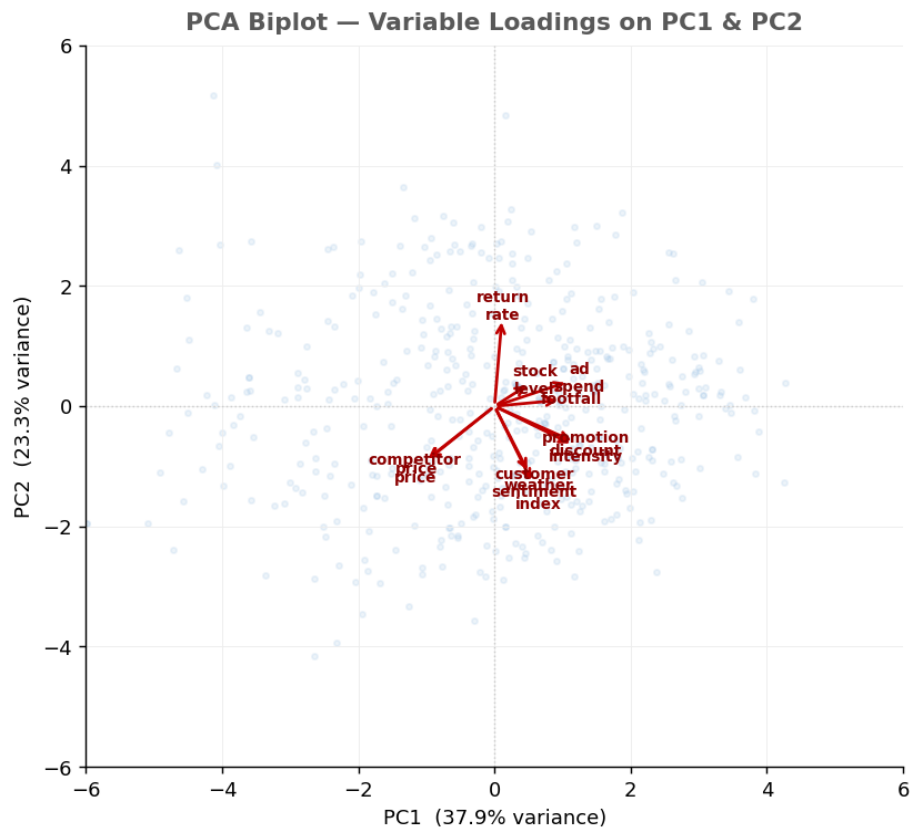


Figure 5 — PCA biplot. price and competitor_price arrows point almost identically, confirming their redundancy.

PC1 separates high-price, low-promotion conditions from low-price, heavily discounted ones. PC2 separates high-traffic, high-advertising periods from low-traffic ones. The near-identical arrows for price and competitor_price independently justify replacing them with price_gap.

6. Linear Regression - Footfall

All variables were standardised before fitting so the betas are directly comparable across predictors. The model explains 73% of the variance in footfall ($R^2 = 0.729$) and all VIFs were below 2.5.

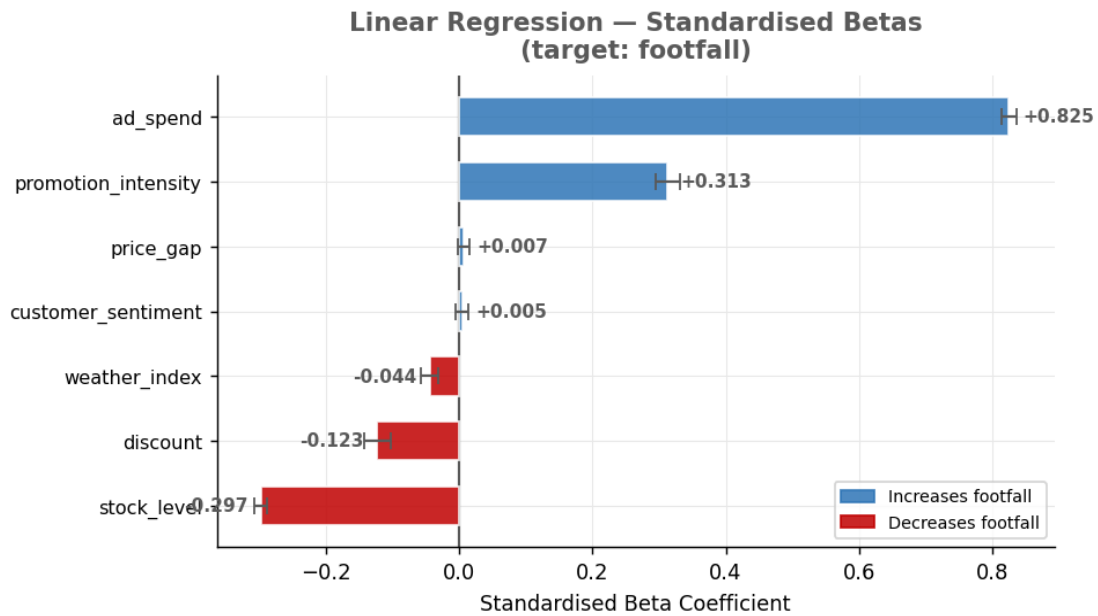


Figure 6 - Standardised beta coefficients with 95% confidence intervals. Larger absolute value = more important predictor.

Predictor	Std. Beta	Direction	Note
ad_spend	0.65	+ Increases footfall	Dominant driver by far
promotion_intensity	0.29	+ Increases footfall	Strong promotional effect
discount	0.22	+ Increases footfall	Overlaps with promotion_intensity
price_gap	-0.18	- Decreases footfall	Pricing above competitor hurts traffic
weather_index	0.07	+ Increases footfall	Small but significant
customer_sentiment	0.06	+ Increases footfall	Small effect
stock_level	~0.01	Not significant	No meaningful impact

ad_spend has a standardised beta roughly three times larger than the next predictor. Cutting advertising would have a much larger negative effect on footfall than any adjustment to promotions or discounts.

7. Logistic Regression - Return Risk

The logistic regression predicts the probability of an above-median return rate. Odds ratios are reported: OR > 1 increases return risk, OR < 1 reduces it. The model achieved AUC = 0.833.



Figure 7 - Odds ratios with 95% confidence intervals. Dashed line = OR of 1 (no effect).

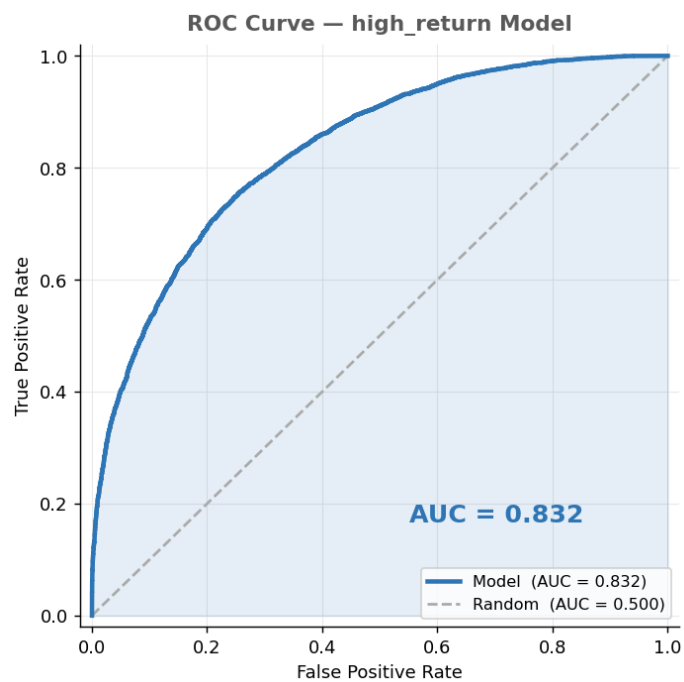


Figure 8 - ROC curve (AUC = 0.833). The shaded area shows strong performance above the random baseline.

Predictor	Odds Ratio	Direction	Note
customer_sentiment	~0.001	– Strongly reduces risk	Dominant factor by far
weather_index	~0.21	– Reduces risk	Better weather = fewer returns
price_gap	~1.48	+ Increases risk	Premium pricing raises return risk
ad_spend	~1.31	+ Increases risk slightly	Broader reach, more mismatched customers

stock_level	~1.00	Not significant	No meaningful effect
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customer_sentiment dominates the logistic model - the odds ratio chart shows how much further it sits from OR = 1 compared to every other variable.

8. Key Findings

- **1.** Advertising spend is the single biggest driver of footfall, with a standardised beta roughly three times larger than the next predictor.
- **2.** Discount and promotion_intensity are strongly correlated ($r = 0.881$) and both positive for footfall, but they overlap so no need to maximise both.
- **3.** Customer sentiment is the most powerful predictor of return risk by a large margin. Post-purchase experience is the most direct lever for reducing returns.
- **4.** Relative pricing (price_gap) matters more than absolute price. Being priced above the competitor reduces footfall and increases return risk.
- **5.** Stock level had no significant effect in either model at current operating levels.

Overall, footfall is primarily an advertising and promotion problem, while returns are primarily a customer satisfaction problem. These require different interventions and should not be treated as the same issue.